

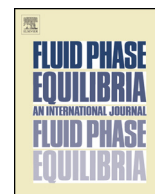


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New modified UNIFAC parameters using critically evaluated phase equilibrium data[☆]



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ABSTRACT

New modified UNIFAC property prediction model parameters are reported for 89 main groups and 984 group–group interactions using critically evaluated phase equilibrium data including vapor–liquid equilibrium (VLE), liquid–liquid equilibrium (LLE), solid–liquid equilibrium (SLE), excess enthalpy (HE), infinite dilution activity coefficient (AINF) and excess heat capacity (CPE) data. The new algorithmic framework for quality assessment of phase equilibrium data is applied for qualifying the consistency of data and screening out possible erroneous data. Substantial improvement over previous versions of UNIFAC is observed due to inclusion of experimental data from recent publications and proper weighting based on a quality assessment procedure. The systems requiring further verification of phase equilibrium data were identified where insufficient number of experimental data points is available or where existing data are conflicting.

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1. Introduction

Various versions of UNIFAC models [1–10] are commonly used for process design when experimental determination of phase equilibrium data is difficult or time consuming. In our previous publication [11], a new procedure for evaluation of UNIFAC interaction parameters was proposed and complete set of group–group interaction parameters was provided for 52 main groups. Group–group interaction parameters were reported for NIST-UNIFAC and NIST-KT-UNIFAC models. Recently, an algorithmic framework for quality assessment of phase equilibrium data [12,13] which covers VLE, LLE, SLE, ACT and HE data has been developed and implemented in the NIST ThermoData Engine [14–23] designed to be a software realization of the concept of dynamic data evaluation for thermophysical and thermochemical properties [24,25]. Implementation of this algorithmic framework can be used to assess phase equilibrium data and screen out

possible erroneous data sets. Furthermore, this procedure can be applied for the development of UNIFAC model parameter matrices.

Because of the requirements of having additional parameters associated with the temperature-dependent terms, the modified UNIFAC model proposed by Gmehling and co-workers [2–6] covers wide range of phase equilibrium data including HE and SLE. Some of the parameters of this model are published in the literature whereas many others are available to the UNIFAC Consortium [26] members only. As the amount of published phase equilibrium data grows continuously, there is a need for periodic updates of the parameter matrix which reflects most recent information for phase equilibrium. In this work, we present the results of our assessment of the interaction parameters for modified UNIFAC (NIST-modified UNIFAC model) using data set prescreening and weighting procedures based on the software implementation of the algorithmic framework for quality assessment of phase equilibria data we reported previously [12,13].

2. Description of NIST-modified UNIFAC model and parameter fitting procedure

2.1. NIST-modified UNIFAC models

The expressions for activity coefficient and other properties using various versions of UNIFAC models are presented in the

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previous publications [1–10], and details will not be discussed here. The modified UNIFAC model proposed by Gmehling and co-workers [2–6] has a slightly different formulation for average volume parameters. The values used are

$$\phi_i = \frac{r_i}{\sum_{j=1}^{NC} x_j r_j} : \text{used in all UNIFAC versions} \quad (1)$$

$$\phi'_i = \frac{r_i^{3/4}}{\sum_{j=1}^{NC} x_j r_j^{3/4}} : \text{used in modified UNIFAC together with } \phi_i \quad (2)$$

where r_i is volume of i -th structural subgroup and x_i is mole fraction among all subgroups composing the mixture.

Interaction parameters in original UNIFAC [1] were assumed as independent in temperature.

$$\Delta u_{mk} = a_{mk,1} \quad (3)$$

where $a_{mk,1}$ is the temperature independent coefficient of parameter for m – k group interaction.

For the improvement in the temperature dependencies in VLE and estimation of excess enthalpies, temperature (T) – dependent formulations can be used. The first order form with regard to temperature was used in KT-UNIFAC [8] and the second order form was used in modified UNIFAC [2].

$$\Delta u_{mk} = a_{mk,1} + a_{mk,2}(T - T_0) \quad (4)$$

For KT-UNIFAC, T_0 is a constant, and for NIST-KT-UNIFAC [11] $T_0 = 0$. For modified UNIFAC,

$$\Delta u_{mk} = a_{mk,1} + a_{mk,2}T + a_{mk,3}T^2. \quad (5)$$

Exactly the same formulations previously described for modified UNIFAC were used for the development of the NIST-modified UNIFAC matrix.

2.2. Assessment of phase equilibrium data

In our previous publications, we described a quality assessment algorithm for vapor–liquid equilibrium data (VLE) [12] and a hierarchical procedure for checking other types of phase equilibrium data (LLE, SLE, ACT and HE) [13]. It is of a particular importance, the use of the quality factors for VLE data sets to assess their weighting parameters “blending” them together for the construction of the UNIFAC matrix. Also, quality factors developed for solid–liquid equilibrium (SLE) data sets were used for weighting in the adequacy function described in the next section. For HE, ACT and LLE, the erroneous data sets were excluded from consideration using the procedure outlined previously [13].

2.3. Parameter fitting procedure

The adequacy (objective) function for the regression of UNIFAC parameters is given as follows;

$$A_{\text{total}} = w_{\text{VLE}}A_{\text{VLE}} + w_{\text{LLE}}A_{\text{LLE}} + w_{\text{SLE}}A_{\text{SLE}} + w_{\text{AC}}A_{\text{AC}} + w_{\text{HE}}A_{\text{HE}} + w_{\text{CPE}}A_{\text{CPE}} \quad (6)$$

where $A_{\text{VLE}}, A_{\text{LLE}}, A_{\text{SLE}}, A_{\text{AC}}, A_{\text{HE}}$ and A_{CPE} are the adequacy functions for VLE, LLE, SLE, activity coefficients, excess enthalpy, and excess heat capacities, respectively, and w represents a weighting factor associated with each quantity.

Detailed discussion of the adequacy function and inclusion of uncertainty associated with the data sets are described in our previous publication [11]. For LLE, SLE and CPE which were not presented in [11], a similar approach was used. The weighting factors of each type of data can be adjusted to give better prediction capabilities of certain data types. In this work, highest priority was given to the VLE data and less weight was given to other types of phase equilibrium data.

Because of the principal limitations of the local composition model [27], the modified UNIFAC model cannot successfully predict VLE, SLE and LLE simultaneously except for a few cases. This specific version of parameters is intended to provide the best predictive capabilities for VLE, and we are planning to develop separate sets of parameter matrices optimized for SLE and LLE in future work.

Both the maximum-likelihood method proposed by Anderson et al. [28] and simplex method [29] were used for the fitting of group–group interactions parameters. Due to instability problems arising from derivative evaluation in the maximum-likelihood method, the robust simplex method was used in many cases.

Obviously, generation of all parameters simultaneously in one step is impractical because of the enormity of the optimization space involving hundreds of parameters. Therefore, the fitting algorithm represented a stepwise process. First, each group–group interaction parameter was determined based on the analysis of data for a given class of components (i.e.; alkane + benzene for $\text{CH}_2 + \text{ACH}$ interaction, alkane + alcohol for $\text{CH}_2 + \text{OH}$ interaction, etc. . . .). After completion of this first step, each interaction was re-examined and “tuned-up” a number of times within the whole set of phase equilibrium data related to this specific interaction to produce more accurate results.

3. Results and discussion

3.1. Status of NIST-modified UNIFAC parameter matrix

Experimental phase equilibria data collected within the NIST SOURCE Data Archival System [30] and processed through NIST ThermoData Engine [14] were used in the parameter optimization process. After initial data prescreening conducted in accordance to the procedures described in [13], 186,773 VLE data points, 15,662 LLE data points, 7943 SLE data points, 15,662 ACT data

Table 1
Performance comparison of various UNIFAC models.^a

Model	VLE		LLE	SLE	HE	ACT	CPE
	Δp (%)	Δy (%)	Δx (%)	Δx (%)	ΔH (%)	$\Delta \gamma$ (%)	ΔCp (%)
Original UNIFAC [32]	7.21	2.06	532.95	43.48	240.07	185.88	292.32
Modified UNIFAC [2–6]	4.83	1.81	74.96	42.81	112.12	32.94	555.18
NIST-modified UNIFAC (present article)	3.80	1.76	71.40	32.22	143.16	24.98	230.51
No. of data points	46,258	26,409	6,010	2,712	36,601	5,619	1,446

^a Model performance assessments are limited to components composed of common groups (main group 1–40) among three versions of UNIFAC. Systems with unavailable parameters are excluded from the analysis.

points, 117,908 HE data points, and 5303 CPE data points were used in the parameter fitting. A total 349,251 experimental data points were used to obtain 984 group–group interaction parameters.

89 main-groups and 208 sub-groups for the NIST-modified UNIFAC model are shown in Table 1 of the Supplementary material. In order to preserve compatibility with previous UNIFAC versions, we use the same main group numbers and sub-group numbers where possible. Also, the same values of Q_k and R_k values for sub-groups were used when available [2–6]. For unpublished values, we pre-estimated values for Q_k and R_k using Bondi's method [31] and then further optimized the resulting values to provide the best fitting possible.

Current status of group–group interaction parameters for NIST-modified UNIFAC is summarized in Fig. 1. Values of the parameters are listed in Table 2 of the Supplementary material. In total, 984 group–group interaction parameters are reported (total number of the parameters previously published for modified UNIFAC model [2–6] covered 659 group–group interactions). Due to the second order nature of the temperature dependency of the parameters, extrapolation outside the fitted range may produce unexpected results. The parameters should be used inside the specified temperature ranges. Parameters are subject to change whenever new experimental data to be published, and we plan to make parameter updates available without any restrictions on the web site of the Thermodynamics Research Center at NIST (<http://trc.nist.gov>).

3.2. Examples for new main groups and sub-groups

Most of the main groups defined in modified UNIFAC [2–6] are included in NIST-modified UNIFAC, and several new groups are introduced for improved calculations. Thus, previously deployed versions of UNIFAC for mixtures containing components with

multiple hydroxyl groups (–OH groups) attached to the adjacent carbon atoms showed insufficient predictive capabilities compared to those for other alcohols. The main reason for such behavior is due to reduction in hydrogen bonding interaction due to steric hindrances. For that reason glycerol (which has three hydroxyl groups attached to the adjacent carbon atoms) was treated as a new molecular group (main group 96, subgroup 206). Examples of improvements resulting from this more detailed group classification are presented in Fig. 2 (glycerol + water mixture) and Fig. 3 (glycerol + methanol mixture).

Previously, in modified UNIFAC, ethylene glycol was treated as a molecular group, however, that is not applicable for components such as 1,2-propanediol and 2,3-butanediol. Five new sub-groups are now defined for diols with hydroxyl groups attached to the adjacent carbon atoms. With the introduction of new groups, the calculations were significantly improved as shown in Fig. 4.

Size parameters (R_k and Q_k) for CF2 groups (main group 40) were slightly adjusted to yield better results for mixtures containing fluorohydrocarbons. Improvements are illustrated in Fig. 5.

3.3. Examples of improvements associated with updated experimental data

The number of reported experimental thermophysical and thermochemical property data points has doubled within the last decade [37] while no publicly available updates for the modified UNIFAC matrix [2–6] have been reported in the last 8 years. The impact of the updated experimental data on the parameter improvements is illustrated in Figs. 6–8.

Fig. 6 shows VLE calculations for acrylonitrile + *n*-hexane system. As follows from the results illustrated on Fig. 6, NIST-modified UNIFAC provides better predictive capabilities in

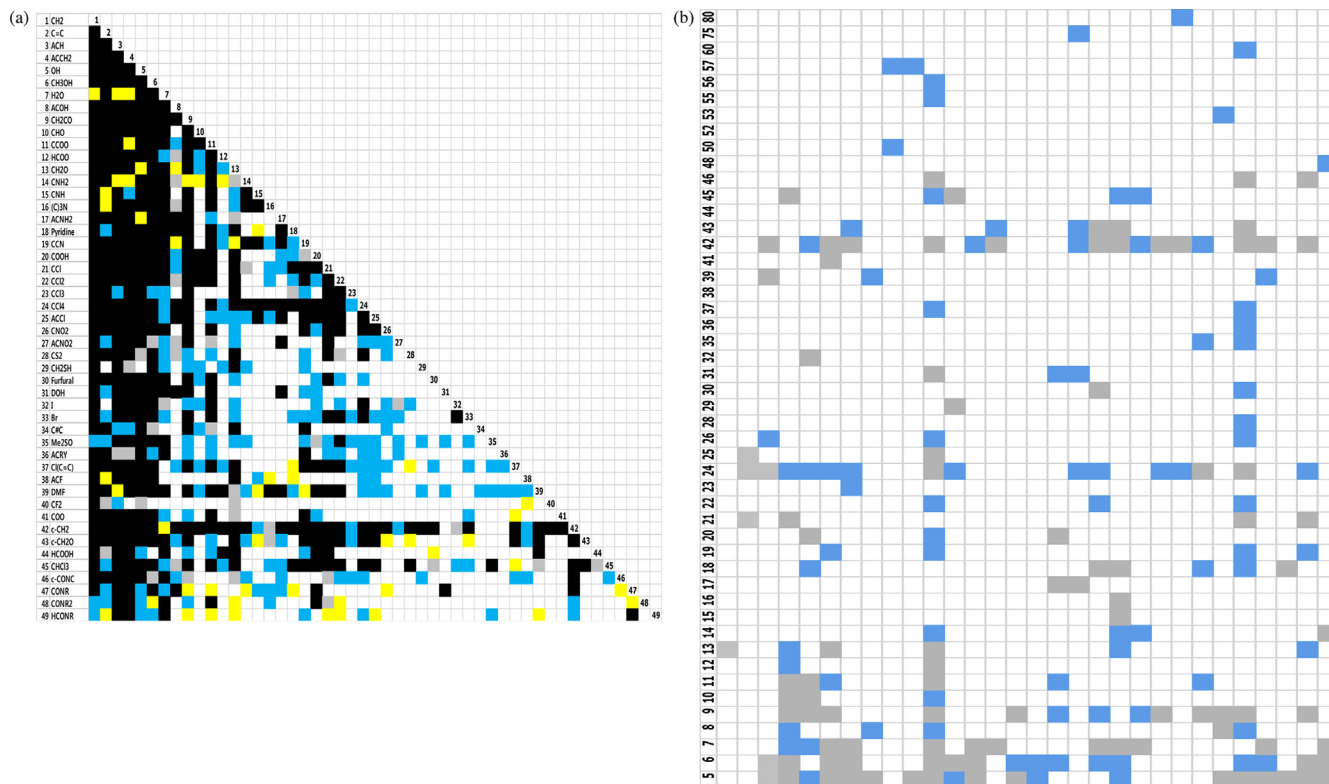


Fig. 1. Current status of the NIST-modified UNIFAC group–group interaction parameter matrix. (black-updated parameters, gray-newly estimated parameters, blue-parameters from insufficient data or conflicting data, yellow-original parameters retained without change). Main groups and subgroups are defined in Table 1 of the Supplementary material and values of parameters are listed in Table 2 of the Supplementary material. (a) Status of parameters for previously published [2–6] main groups. (b) Status of parameters for unpublished main groups.

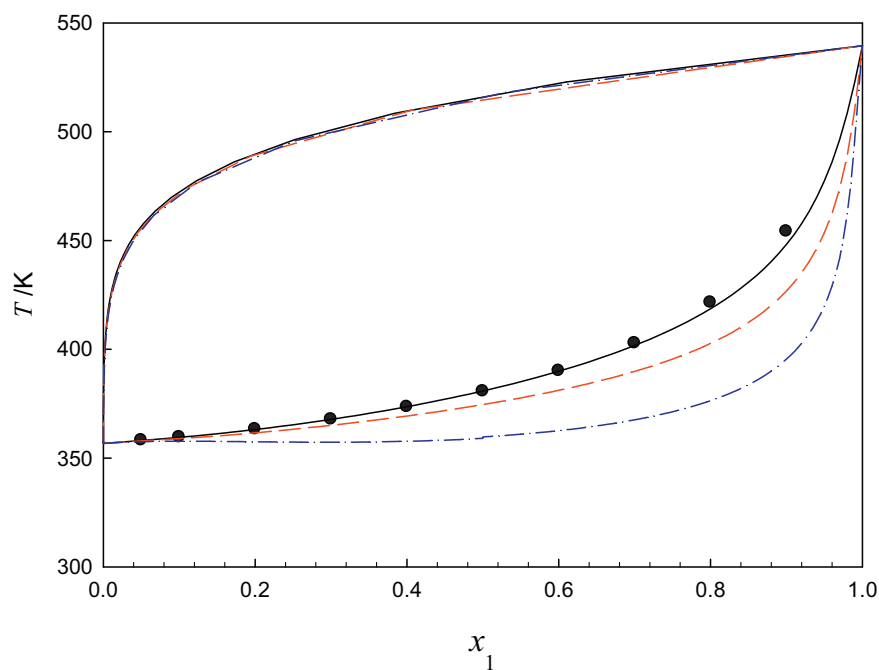


Fig. 2. Comparison of experimental data with UNIFAC predictions for the system (glycerol + water) at pressure $p = 54.72$ kPa. Solid lines (black) are NIST-modified UNIFAC calculations, dashed lines (red) are original UNIFAC [32] calculations, and dash-dot lines (blue) are modified UNIFAC [2–6] calculations, the source of experimental data is Soujanya et al. [33] (●). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

comparison with two other versions of UNIFAC due to the fact that the experimental data [38] published in 1996 have not previously been used for the modified UNIFAC parameter matrix update. Fig. 7 provides similar illustration for the system acetic acid + *n*-methyl acetamide while Fig. 8 shows the results of such a comparison for 1,1,2,2-tetrachloroethane + tetrahydrofuran system.

3.4. Examples with conflicting and insufficient data sets

It is important to emphasize that conflicting or insufficient experimental data sets represent a major factor hindering progress in further improvement of the predictive capabilities of the UNIFAC-based methods. For an example, (propionic acid + carbon

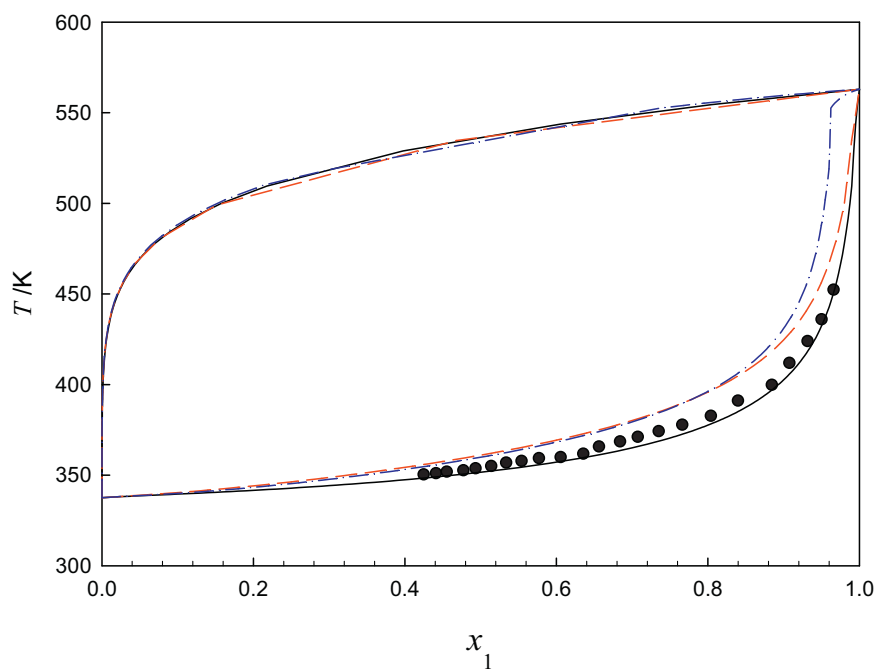


Fig. 3. Comparison of experimental data with UNIFAC predictions for the system (glycerol + methanol) at pressure $p = 101$ kPa. Solid lines (black) are NIST-modified UNIFAC calculations, dashed lines (red) are original UNIFAC [32] calculations, and dash-dot lines (blue) are modified UNIFAC [2–6] calculations, the source of experimental data is Oliveira et al. [34] (●). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

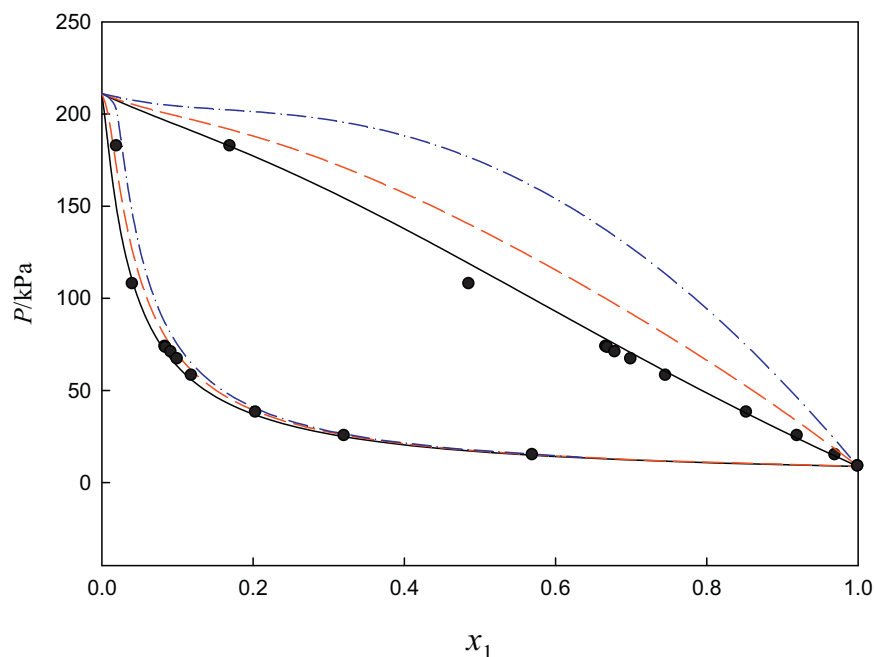


Fig. 4. Comparison of experimental data with UNIFAC predictions for the system (1,2-propylene glycol + water) at temperature $T = 395.1$ K. Solid lines (black) are NIST-modified UNIFAC calculations, dashed lines (red) are original UNIFAC [32] calculations, and dash-dot lines (blue) are modified UNIFAC [2–6] calculations, the source of experimental data is Lancia et al. [35] (●). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

tetrachloride) VLE data sets are very important sources for evaluation of (COOH + CCl₄) main group interactions. However, as shown in Fig. 9, experimental data sets available for this system at almost identical conditions [41,42] differ very significantly. Even though quality assessment factor for VLE (Q_{VLE}) may discriminate the quality of data, the evaluated values of Q_{VLE} are very small for both data sets in this case. Similarly, Fig. 10 illustrates conflicting experimental data sets [43–45] for the system (trichloroethylene + furfural).

In many instances group–group interactions were obtained from one available experimental VLE data set. In such cases, the quality assessment procedures are of very limited use because this data set is the only one to contribute to the adequacy function. For that purpose, on Fig. 1 we specifically identified parameters obtained on the basis of small number of data points (less than 20 points) or when available experimental data sets are conflicting. This information can be used by experimentalists to build experimental plans for phase equilibria measurements with the

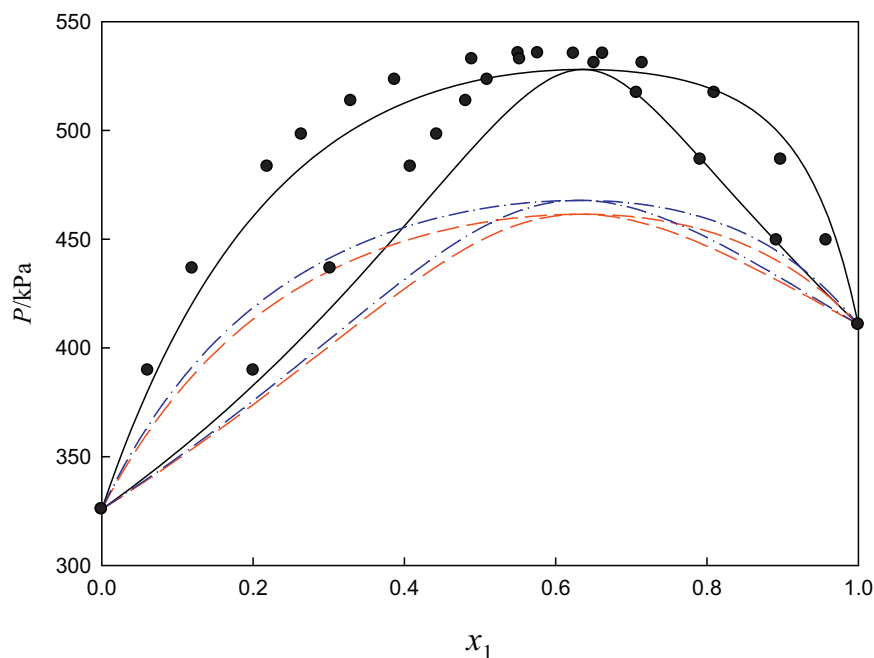


Fig. 5. Comparison of experimental data with UNIFAC prediction for the system (isobutane + 1,1,1,3,3,3-hexafluoropropane) at temperature $T = 303.68$ K. Solid lines (black) are NIST-modified UNIFAC calculations, dashed lines (red) are original UNIFAC [32] calculations and dash-dot lines (blue) are modified UNIFAC [2–6], the source of experimental data is Bobbo et al. [36] (●). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

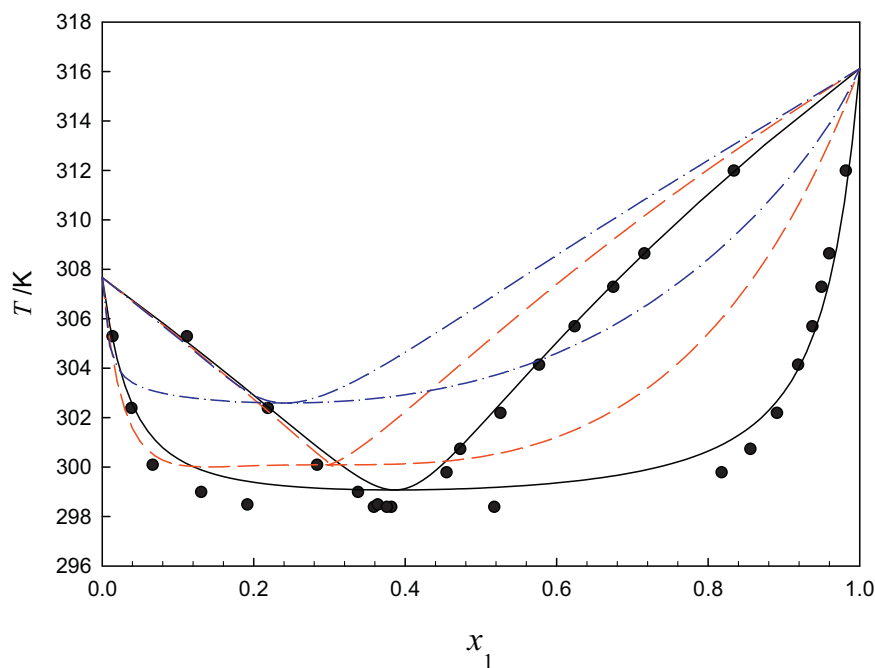


Fig. 6. Comparison of experimental data with UNIFAC predictions for the system (acrylonitrile + *n*-hexane) at pressure $p = 30.04$ kPa. Solid lines (black) are NIST-modified UNIFAC calculations, dashed lines (red) are original UNIFAC [32] calculations, and dash-dot lines (blue) are modified UNIFAC [2–6] calculations, the source of experimental data is de Haan et al. [38] (●). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

purpose of further expansion of the UNIFAC-based predictive methods.

3.5. Extrapolation challenge

As experimental data for a pair of UNIFAC groups are often available in a narrow temperature range (Fig. 11), the use of

extrapolation is commonly desired. However, if extrapolation is undertaken, the testing of the extrapolation behavior (temperature dependence of the calculated properties) is critical. The example of unreliable extrapolation of the infinite dilution activity coefficient of 1,2-dichlorobenzene in its mixture with 5,8,11,14-tetraoxaocadecane is illustrated in Fig. 12. Interaction parameters for this correlation include alkane group CH_2 (group 1), aromatic CH group

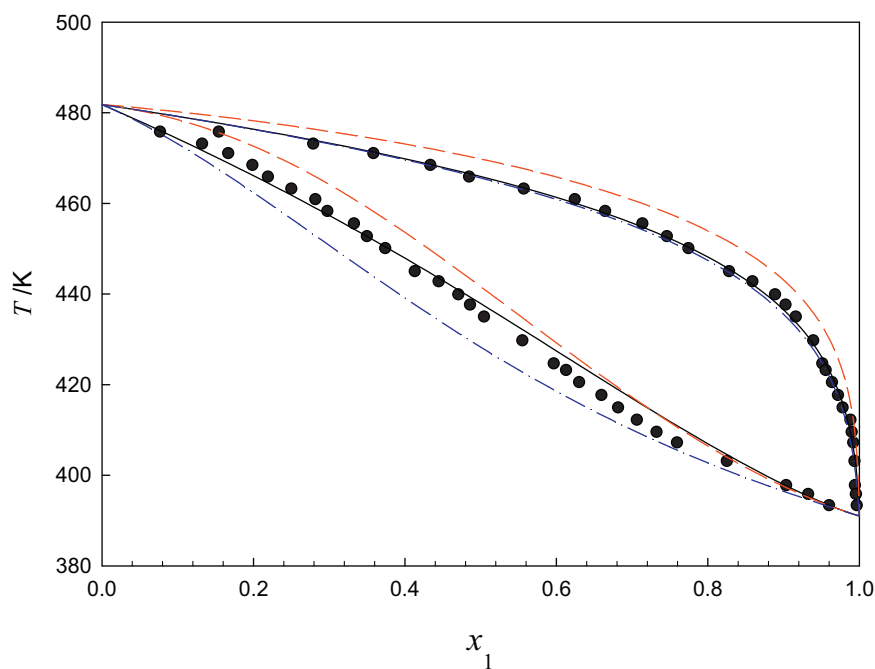


Fig. 7. Comparison of experimental data with UNIFAC predictions for the system (acetic acid + *N*-methylacetamide) at pressure $p = 101.33$ kPa. Solid lines (black) are NIST-modified UNIFAC calculations, dashed lines (red) are original UNIFAC [32] calculations, and dash-dot lines (blue) are modified UNIFAC [2–6] calculations, the source of experimental data is Chang et al. [39] (●). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

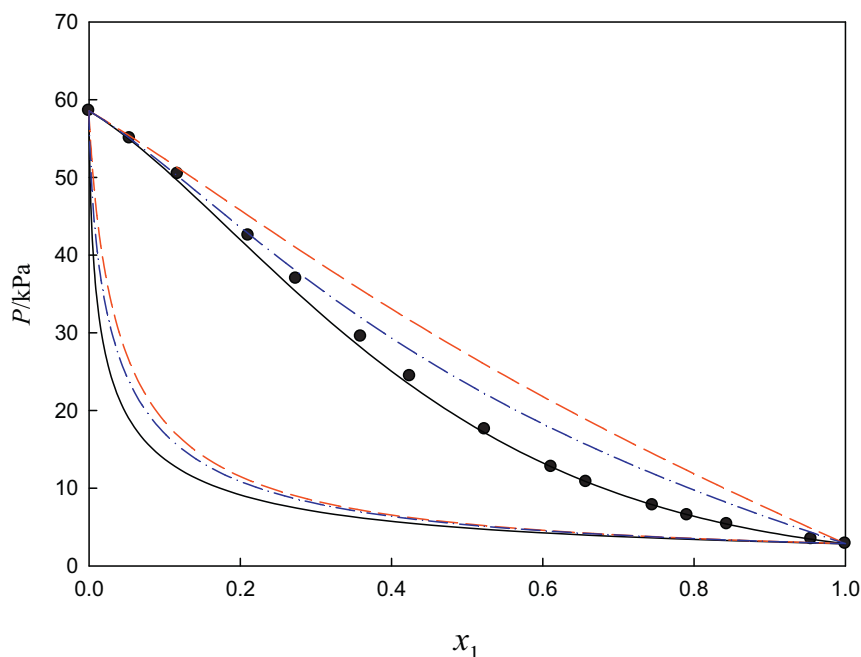


Fig. 8. Comparison of experimental data with UNIFAC predictions for the system (1,1,2,2-tetrachloroethane + tetrahydrofuran) at temperature $T = 323.15$ K. Solid lines (black) are NIST-modified UNIFAC calculations, dashed lines (red) are original UNIFAC [32] calculations, and dash-dot lines (blue) are modified UNIFAC [2–6] calculations, the source of experimental data is Garriga et al. [40] (●). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(group 3), OCC group (group 13) and AC–Cl group (group 25), see Tables 1 and 2 of the Supplementary material. Validity temperature ranges for relevant group–group parameters are as follows: 178–463 K (interaction 1–3); 283–473 K (interaction 1–25); 266–423 K (interaction 1–13); 249–413 K (interaction 3–13); 211–399 K (interaction 3–25); and 298–396 K (interaction

13–25). Overlapping validity range for these 6 interaction parameters is 298–396 K. As shown in Fig. 12, extrapolation outside of this range is unreliable.

Three terms in the temperature dependence of parameters for modified UNIFAC model give additional flexibility to describe excess enthalpy or LLE curve shape, but they also can lead to

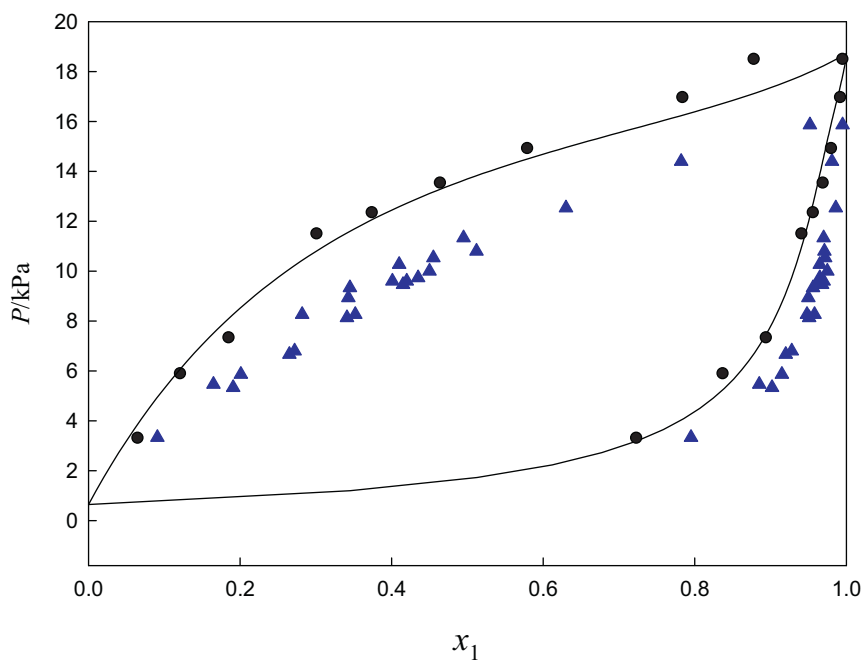


Fig. 9. Example of conflicting data sets available for the mixture (carbon tetrachloride + propionic acid) at almost identical conditions. Solid lines are NIST-modified UNIFAC calculations ($T = 303.14$ K), the sources of experimental data are Tamir et al. [41] ($T = 302.99$ K, (black circles) and Liszi et al. [42] ($T = 303.143$ K, blue triangles). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

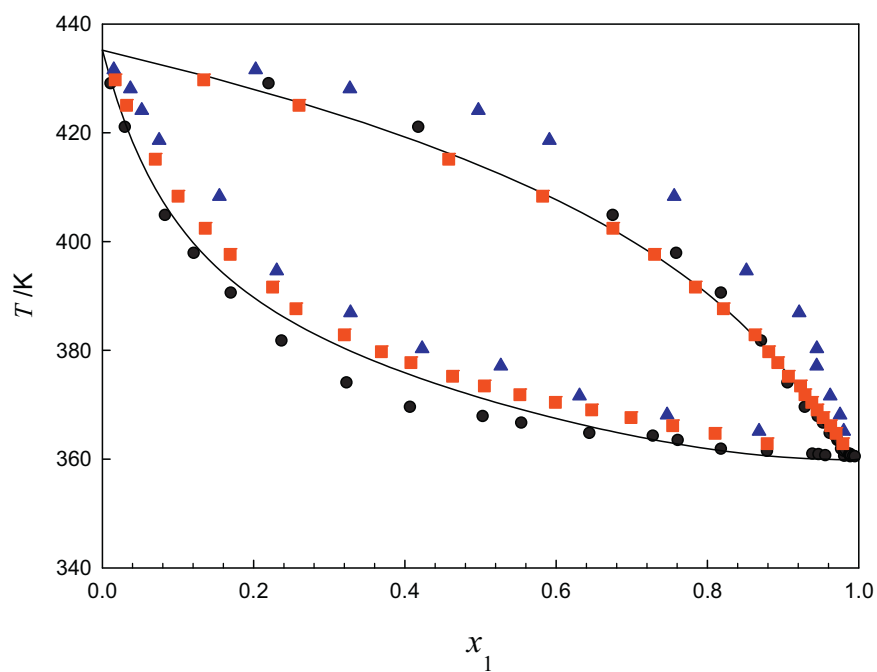


Fig. 10. Example of conflicting data sets for the system (trichloroethylene + furfural) at almost identical conditions. Solid lines are NIST-modified UNIFAC calculations ($P = 101.32$ kPa), the sources of experimental data are: Zivanov et al. [43] ($P = 101.32$ kPa, black circles), Ruchai et al. [44] ($P = 101.32$ kPa, blue triangles), Cabezas et al. [45] ($P = 101$ kPa (red rectangles)). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

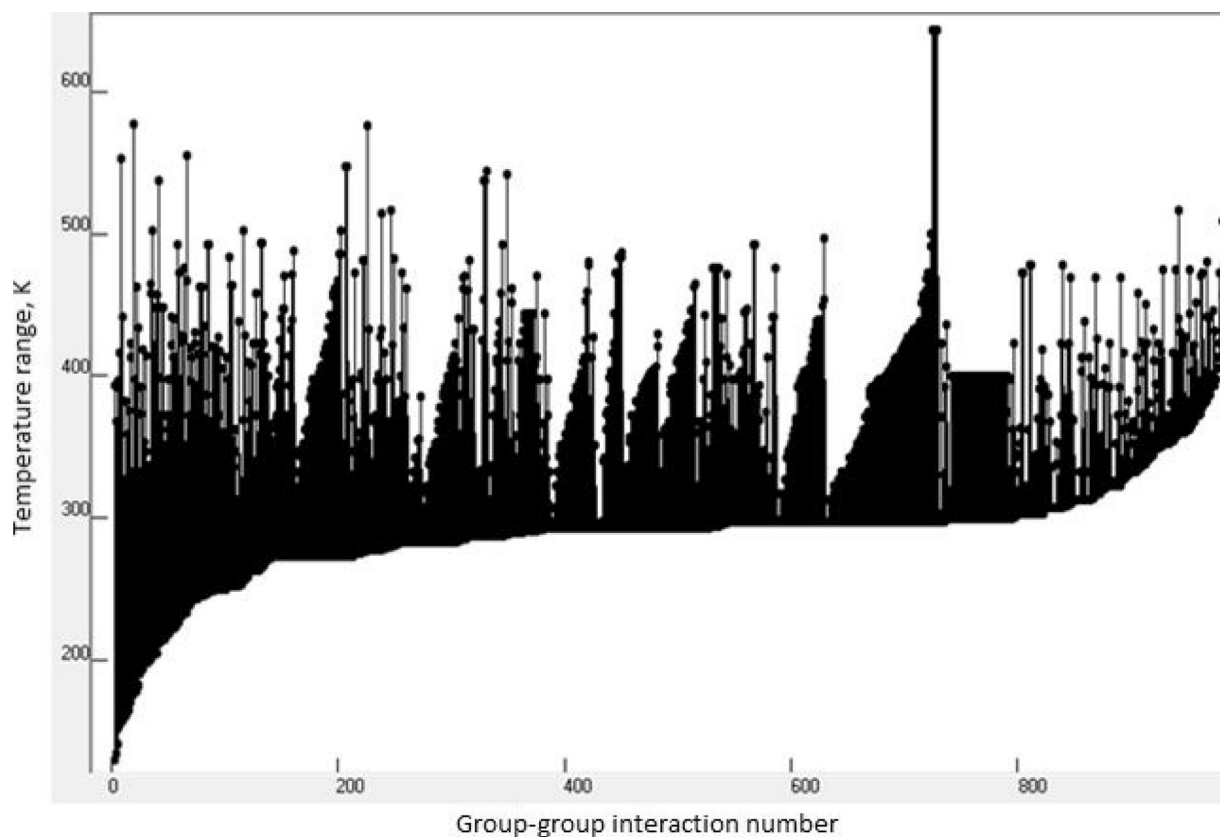


Fig. 11. Temperature ranges of experimental data sets used in the development of the NIST-modified UNIFAC. Group-group interactions are numbered by increasing lower limit of the temperature range, first and upper temperature limit, second, for the experimental data used for parameter regression.

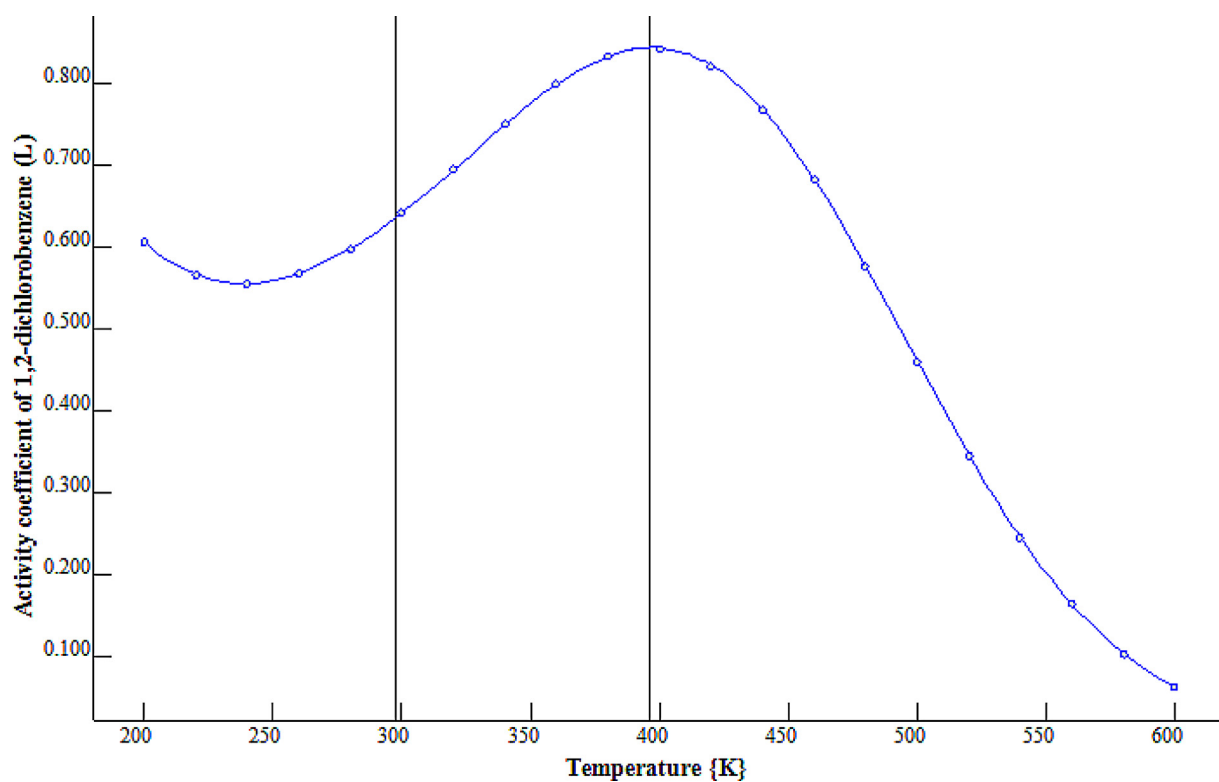


Fig. 12. Example of unreliable extrapolation of the NIST-modified UNIFAC outside of the temperature validity range; infinite dilution activity coefficient as a function of temperature for 1,2-dichlorobenzene in its mixture with 5,8,11,14-tetraoxaocadecane. Overlapping validity temperature range is 298–396 K. The plot was generated with the TDE software [14].

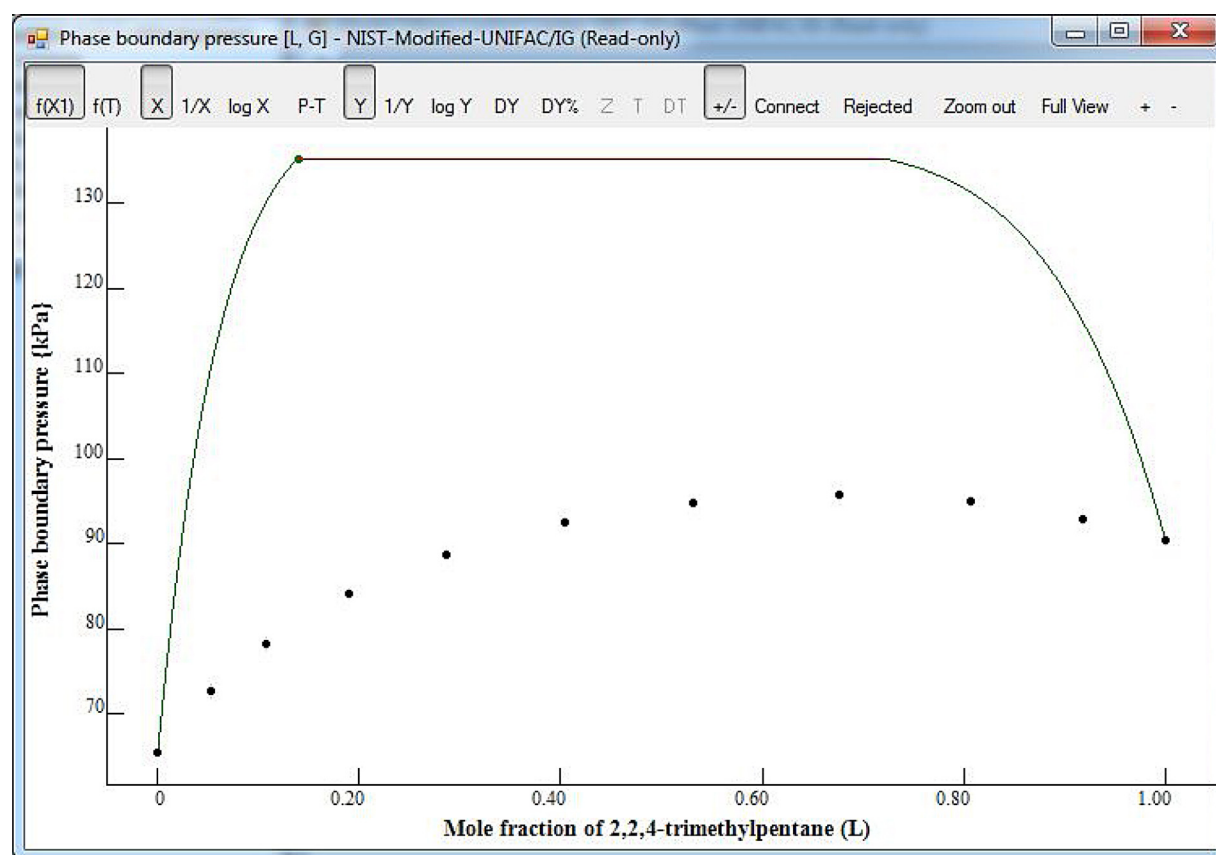


Fig. 13. Example of non-additive VLE behavior for (2,2,4-trimethylpentane + 2,3-dithiabutane) mixture at $T = 368.15$ K. The line represents NIST-modified UNIFAC calculations, the source of experimental data is Sapei et al. [47]. The plot was generated with the TDE software [14]: straight line – predicted LLE region. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

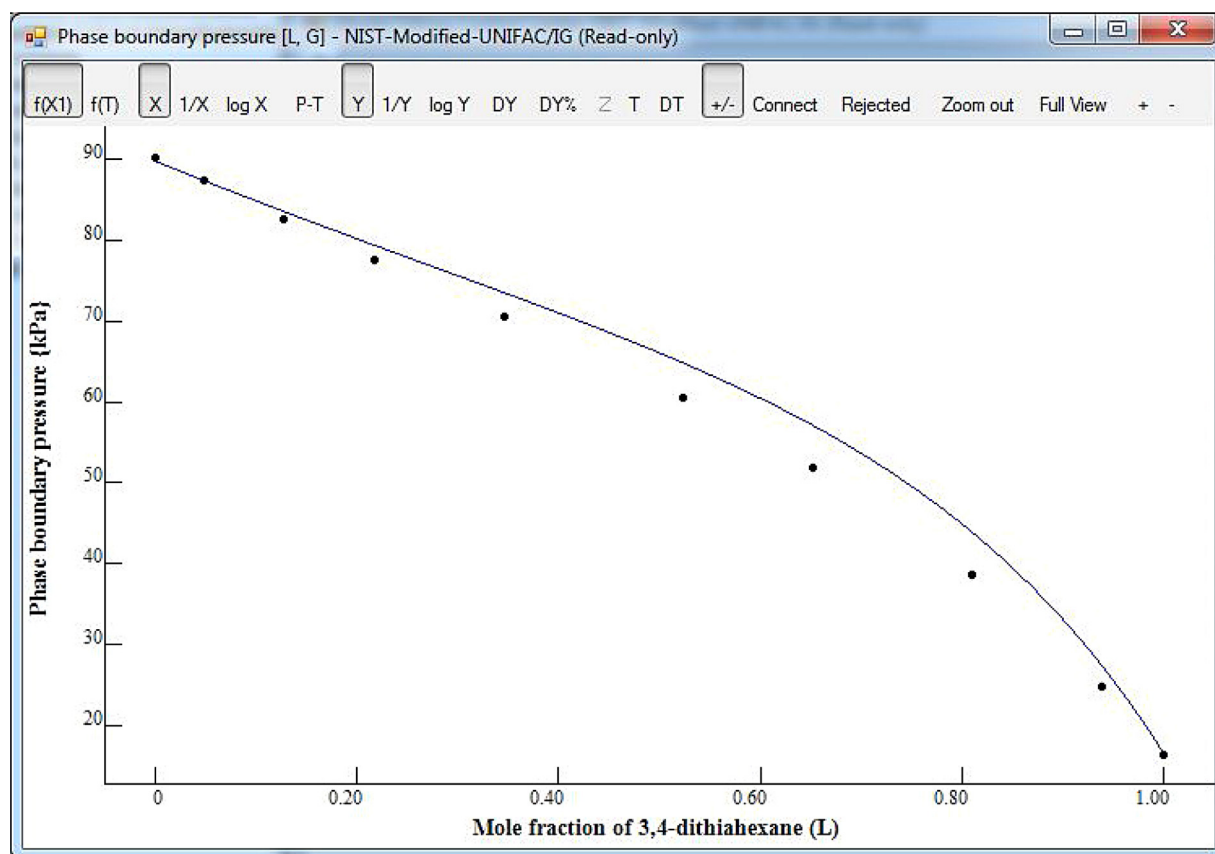


Fig. 14. VLE for (3,4-dithiahexane + 2,2,4-trimethylpentane) mixture at $T = 368.15$ K. The line represents NIST-modified UNIFAC calculation, the source of experimental data is Sapei et al. [47]. The plot was generated with the TDE software [14].

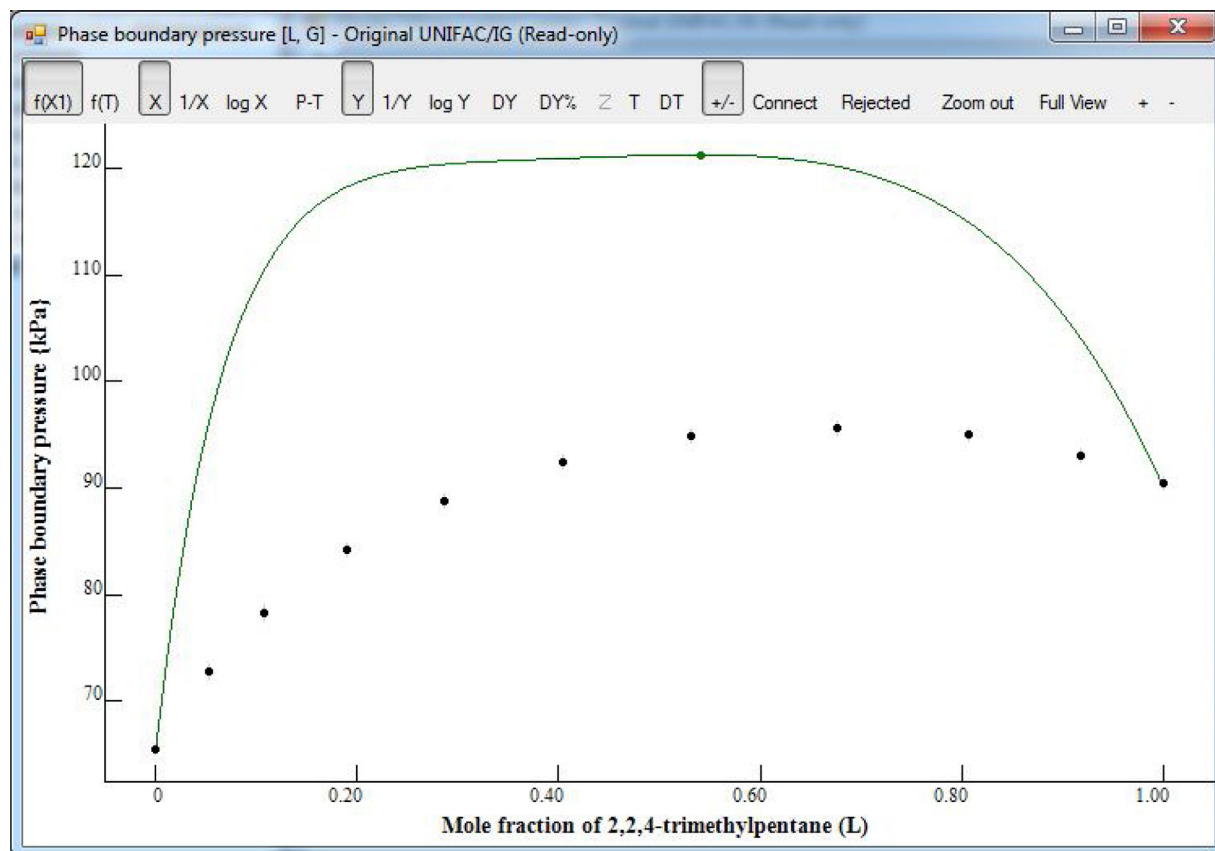


Fig. 15. Example of non-additive VLE behavior for (2,2,4-trimethylpentane + 2,3-dithiabutane) mixture at $T = 368.15$ K. The line represents original UNIFAC (NIST-UNIFAC implementation), the source of experimental data is Sapei et al. [47]. The plot was generated with the TDE software [14].

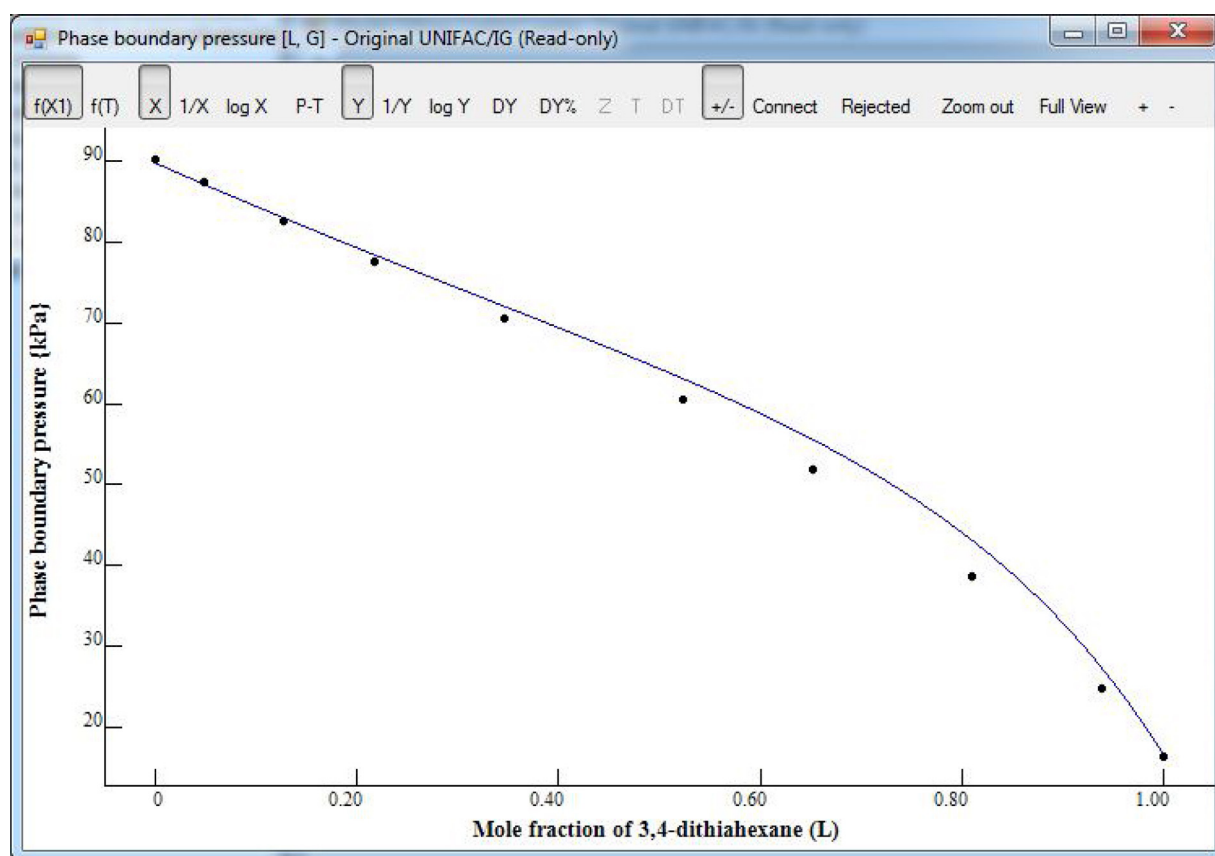


Fig. 16. VLE for (3,4-dithiahexane + 2,2,4-trimethylpentane) mixture at $T = 368.15$ K. original UNIFAC (NIST-UNIFAC implementation), the source of experimental data is Sapei et al. [47]. The plot was generated with the TDE software [14].

“overtraining” that may result in unphysical extrapolation behavior. In order to reduce such a possibility, the use of fitting constraints similar to those deployed in the equation of state fitting [16,46] can be implemented. The constraints for UNIFAC fitting can be either limiting values of the 2nd and 3rd terms, or calculations of the SLE at the eutectic point, as well as elimination of inflections in temperature dependence of activity coefficients. We plan to further study all of these possibilities in the future. In the meantime, as we emphasized previously, the parameters should be used inside the specified temperature ranges.

3.6. Higher-order interactions

Certain strong interactions cannot be described by UNIFAC additivity. In many cases, such as diols or glycerol, special large subgroups are designed ($\text{CH}_2\text{OH}-\text{CH}_2\text{OH}$, $\text{CH}_2\text{OH}-\text{CHOH}-\text{CH}_2\text{OH}$) that consist of several small subgroups. However, the problem is more common as illustrated in case of (2,2,4-trimethylpentane + 2,3-dithiabutane) mixture (Fig. 13). Notably, the structure of 2,3-dithiabutane contains two S-CH₃ subgroups. When those subgroups are separated by CH₂ in (3,4-dithiahexane + 2,2,4-trimethylpentane), the model works much better (Fig. 14). Similar situation exists in original UNIFAC (Figs. 15 and 16). It is unlikely that defining new subgroups can resolve this problem. Introduction of higher-order interactions can be considered as another improvement in the future.

3.7. Overall performance of the NIST-modified UNIFAC

Comprehensive tests showed that NIST-KT-UNIFAC is 44% more accurate than NIST- UNIFAC (over 14,600 binary mixtures tested) and 15% more accurate than NIST-KT-UNIFAC (over 13,700 mixtures tested) for properties other than LLE in terms of weighted RMS (root-mean-square). The weight-averaged RMS for mixture i is calculated as $R_i = 1/N_i \sum_j \sum_k w_{jk} \Delta_{jk}^2$ where w_{jk} is weight of k -th data point in j -th property and Δ_{jk} is its curve deviation. w_{jk} is squared experimental uncertainty, N_i is number of data points for i -th mixture. Model adequacy A is calculated as mean R : $A = \sum_i R_i N_i / \sum_i N_i$. The percent adequacy difference of two models is calculated as $100 \times (A_2 - A_1)$. Special provisions are made for VLE pressure and phase composition: instead of $w_{jk} \Delta_{jk}^2$, summation in the first equation is for $\ln^2 |P_{\text{calc}}/P_{\text{exp}}| (w_{jk} * P_{\text{exp}}^2)$ where P_{exp} and P_{calc} are experimental and calculated property values, respectively. For composition x_1 greater than 0.5 the content of a minor component x_2 is considered as property instead of x_1 .

Comparison of three models (original UNIFAC [32], modified UNIFAC [2–6], and NIST-modified UNIFAC described in present article) was performed for the systems containing only main groups 1 through 40 for systems where parameters were available for all three models. The results of comparison are shown in Table 1. As shown in the table, modified UNIFAC and NIST-modified UNIFAC show similar performances in all aspects of phase

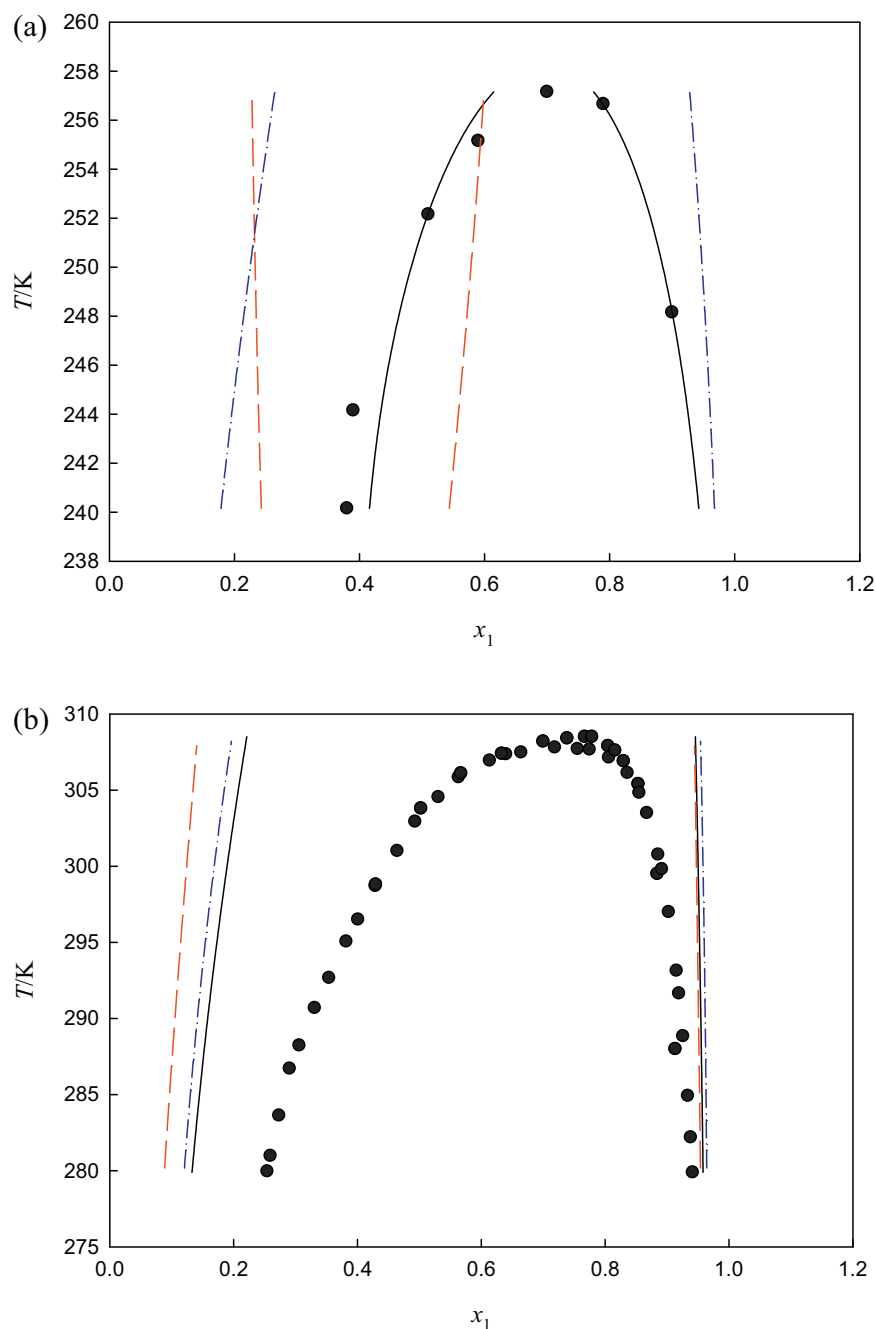


Fig. 17. Performance of various UNIFAC prediction models for liquid–liquid equilibrium (LLE): (a) (nitromethane + 1,3,5-trimethylbenzene) system at atmospheric pressure. The source of experimental data (●) is: Clever et al. [48]; (b) (ethanol + *n*-tetradecane) system at atmospheric pressure. The sources of experimental data (●) are: Dahlman et al. [49] and Matsuda et al. [50]. Solid lines (black) are NIST-modified UNIFAC calculation results, dashed lines (red) are original UNIFAC [32], and dash-dot lines (blue) are modified UNIFAC [2–6]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

equilibrium calculations. These performances are significantly superior of that of original UNIFAC.

Because of the different physical nature of various phase equilibrium properties, it is impossible to reach optimized performance of the UNIFAC models for all of them simultaneously. For example, LLE composition is not directly controlled by either activity coefficients or their temperature derivatives but rather by the inflections of the excess Gibbs energy curve at constant

temperature as a difference to SLE and VLE. That is why the parameter matrix for NIST-modified UNIFAC reported here (see, Supplementary material) was developed with a particular emphasis on VLE, and therefore it is recommended primarily for modeling vapor–liquid equilibria. While for some systems, the properties other than VLE can still be successfully modeled using NIST-modified UNIFAC (see, relevant examples for LLE and HE illustrated in Figs. 17(a) and 18(a), correspondingly), significant

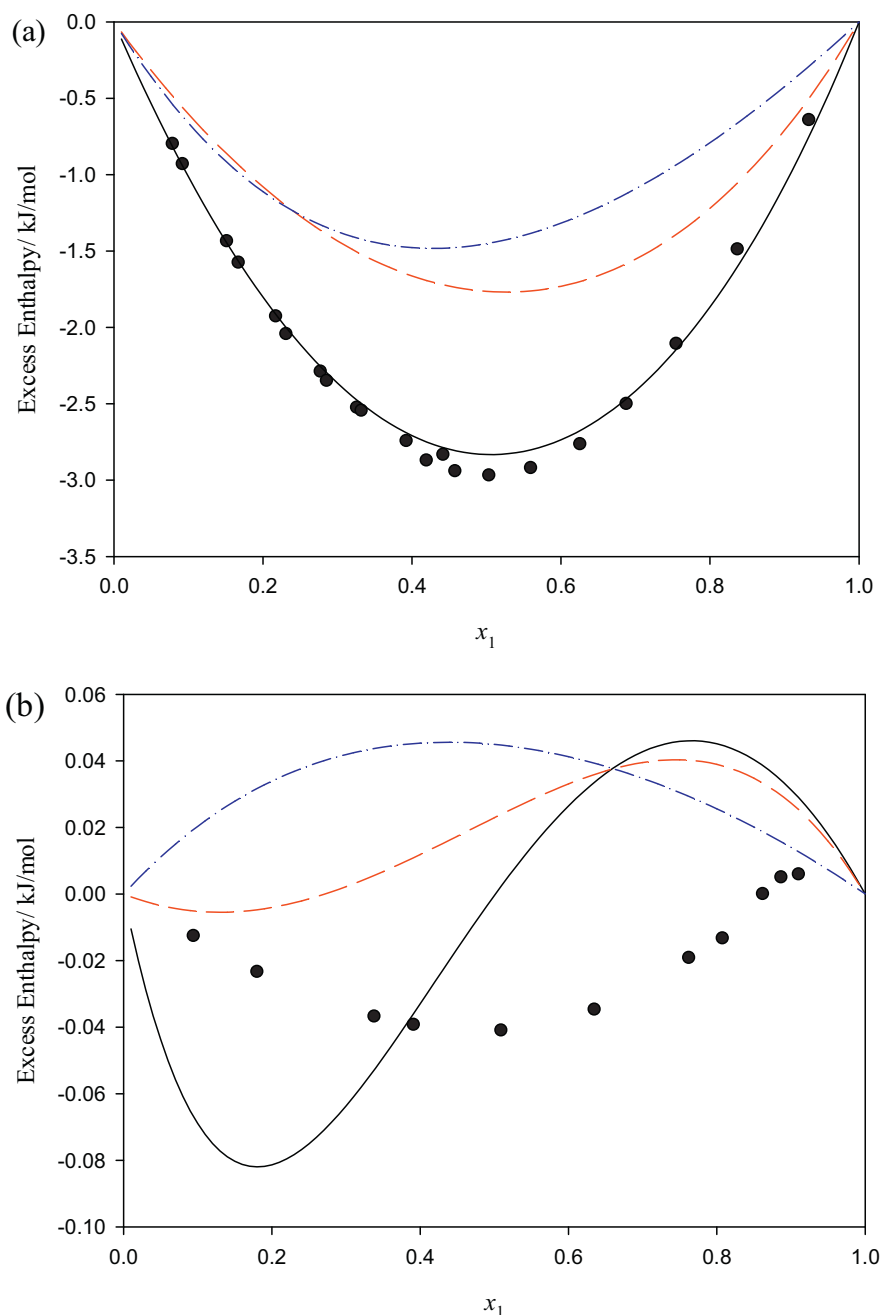


Fig. 18. Performance of various UNIFAC prediction models for excess enthalpy (HE): (a) (aniline + N,N-dimethylformamide) at $T = 298.15$ K and $p = 101.325$ kPa. The source of experimental data (●) is: Ramadevi et al. [51]. (b) (diisopropyl ether + p-xylene) at $T = 308.15$ K and $p = 101.325$ kPa. The source of experimental data (●) is: Mahl et al. [52]. Solid lines (black) are NIST-modified UNIFAC calculation results, dashed lines (red) are original UNIFAC [32], and dash-dot lines (blue) are modified UNIFAC [2–6]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

deviations might be observed for these properties in some other cases (see examples illustrated in Figs. 17(b) and 18(b)).

4. Conclusions

New modified UNIFAC property prediction model parameters are reported for 89 main groups and 984 group–group interactions by using critically evaluated phase equilibrium data including vapor–liquid equilibrium (VLE), liquid–liquid equilibrium (LLE), solid–liquid equilibrium (SLE), excess enthalpy (HE), infinite dilution activity coefficient (AINF) and excess heat capacity

(CPE) data. The new algorithmic framework for quality assessment of phase equilibrium data is applied for qualifying the consistency of data and screening out possible erroneous data. Substantial improvement over previous versions of UNIFAC is observed due to inclusion of experimental data from recent publications and proper weighting based on a quality assessment procedure.

Various options for better extrapolation capabilities will be explored in the near future. Obtained parameters are recommended primarily for correlations of the VLE data. In the future, we also plan to obtain similar matrixes targeting SLE and LLE properties. Some of the identified parameters have been obtained

with the use of conflicting or insufficient experimental data sets. Additional phase equilibrium measurements for systems containing these parameters are recommended.

It is critical, in our opinion, to have assessments available for estimations of the uncertainties associated with the UNIFAC predictions. To this end, currently we assigned uniform variance level to all parameters. In the future, we plan to build a block-diagonal covariance matrix for all UNIFAC parameters where each block (6×6 in case of Modified UNIFAC) will correspond to one interaction (a pair of subgroups) derived from experimental uncertainties and model deviation of available data.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.fluid.2014.12.042>.

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Update

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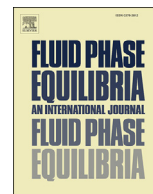
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Corrigendum

Corrigendum to “New modified UNIFAC parameters using critically evaluated phase equilibrium data” [Fluid Phase Equilib. 388 (2015) 128–141][☆]Jeong Won Kang^a, Vladimir Diky^{b,*}, Michael Frenkel^b^a Department of Chemical and Biological Engineering, Korea University, 5-1 Anam-Dong, Sungbuk-Ku, Seoul 136-701, South Korea^b Applied Chemicals and Materials Division, National Institute of Standards and Technology, Boulder, CO 80305-3337, USA

Our paper reporting NIST-modified UNIFAC method contained a few typographical errors, as documented by J. Gmehling [1]. Further review of the original publication and its supplementary materials revealed minor additional necessary corrections. We therefore compiled this corrigendum in order to provide a comprehensive resource of all necessary corrections.

The most important correction is transposing the R and Q columns in Supplement Table 1. The correct header should look like

No.	Sub-group Name	R_k	Q_k	Example
1	CH3	0.6325	1.0608	hexane

2 CH3, 4 CH2

... with all numerical values unchanged.

In addition, main group 86 was erroneously defined as 84. The corresponding line in Supplement Table 1 should be as follows:

No.	Sub-group Name	R_k	Q_k	Example
(86) C2Cl4	C2Cl4	3.3810	3.5845	tetrachloroethene

1 C2Cl4

Main group 83 should be spelled as

(83) ACCO

All other additions and corrections below apply to the examples in Supplement Table 1.

The example for subgroup 39 should be corrected:

39	AC2N	1.0731	0.3530	2,6-dimethylpyridine
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1 AC2N, 3ACH, 2 CH3

DOI of original article: <http://dx.doi.org/10.1016/j.fluid.2014.12.042>.

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<http://dx.doi.org/10.1016/j.fluid.2017.02.014>
0378-3812

The example for subgroup 160 should be corrected:

160	AC2HO	1.1404	0.7600	2,3-benzofuran	5 ACH, 1 AC, 1 AC2HO
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The example for subgroup 162 should be corrected:

162	c-CH-NH	0.9796	0.6240	2,6-lupetidine	1 c-CH-NH, 2 CH3, 3 c-CH2, 1 c-CH
-----	---------	--------	--------	----------------	-----------------------------------

The example for subgroup 163 should be corrected:

163	c-C-NH	0.7521	0.3960	2,2,6,6-tetramethylpiperidine	1 c-C-NH, 4 CH3, 3 c-CH2, 1 c-C
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The example for subgroup 188 should be corrected:

188	c-CH2-NH	1.2070	0.9360	piperazine	2 c-CH2-NH, 2 c-CH2
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The example for subgroup 198 should be corrected:

198	AC2NH	0.8701	0.2120	2,5-dimethylpyrrole	1 AC2NH, 2 ACH, 2 CH3
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The example for subgroup 305 should be corrected:

305	CNO2	2.8870	2.2410	2-methyl-2-nitropropane	3 CH3, 1 CNO2
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The example for subgroup 306 must be changed:

306	ACNH	1.1849	0.7320	diphenylamine	10 ACH, 1 AC, 1 ACNH
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The example for subgroup 309 should be added:

309	CH2(O)2	0.9347	0.7080	dimethoxymethane	2 CH3, 1 CH2(O)2
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The authors are grateful to Prof. Gmehling for his careful reading of the paper and valuable comments.

Reference

- [1] Jürgen Gmehling, *Fluid Phase Equilib.* 398 (2015) 98–100.